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COURSE SYLLABUS

INTRODUCTION TO COMPUTER SCIENCE (CS101)

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COURSE INFORMATION

Course Code: CS101

Course Title: Introduction to Computer Science

Credits: 3

Semester: Fall 2024

Prerequisites: None

INSTRUCTOR INFORMATION

Name: Professor John Smith

Email: john.smith@university.com

Office Hours: Monday & Wednesday 2:00 PM - 4:00 PM

COURSE DESCRIPTION

This course provides a comprehensive introduction to the fundamental concepts of computer science. Students will learn about algorithms, programming basics, data structures, and computational thinking. No prior programming experience is required.

LEARNING OBJECTIVES

Upon completion of this course, students will be able to:

1. Understand basic programming concepts and logic

2. Write simple programs using Python

3. Analyze and solve problems algorithmically

4. Understand data representation in computers

5. Explain fundamental computer science principles

COURSE TOPICS

Week 1-2: Introduction to Computing and History

Week 3-4: Programming Basics and Python

Week 5-6: Variables, Data Types, and Operators

Week 7-8: Control Structures (if-else, loops)

Week 9-10: Functions and Modules

Week 11-12: Lists, Tuples, and Dictionaries

Week 13-14: File Handling and Exception Handling

Week 15: Introduction to Object-Oriented Programming

Week 16: Final Review and Exam

TEXTBOOKS AND RESOURCES

Required Textbook:

- "Computer Science: An Overview" by Glenn Brookshear
- "Python Crash Course" by Eric Matthes

Online Resources:

- Python.org official documentation
- W3Schools Python Tutorial
- Codecademy Python Course

ASSESSMENT METHODS

Assignment 1: Programming Basics (10%)

Assignment 2: Control Structures (10%)

Assignment 3: Functions (10%)

Assignment 4: Data Structures (10%)

Midterm Exam (25%)

Final Project (15%)

Final Exam (20%)

GRADING SCALE

A: 90-100%

B: 80-89%

C: 70-79%

D: 60-69%

F: Below 60%

KEY CONCEPTS COVERED

- Algorithms
- Variables and Data Types
- Conditional Statements
- Loops (for, while)
- Functions
- Arrays/Lists
- File I/O
- Exception Handling
- Object-Oriented Programming
- Recursion
- Sorting Algorithms
- Search Algorithms

IMPORTANT DATES

Assignment 1 Due: Week 3

Assignment 2 Due: Week 5

Midterm Exam: Week 8

Assignment 3 Due: Week 10

Assignment 4 Due: Week 12

Final Project Due: Week 15

Final Exam: Week 16

COURSE POLICIES

- Attendance is mandatory
- Late submissions will be penalized 10% per day
- Academic integrity strictly enforced
- Collaboration allowed on assignments with proper attribution

CONTACT INFORMATION

Department: Computer Science

Office: Building A, Room 201

Phone: (555) 123-4567